

The analytic $d = 2$ cutoff theorem with unequal starting exponents

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Abstract

The analytic adapter of unit 115 generalised to independent starting exponents $p_1 = (h_1 + 1)/k_1$, $p_2 = (h_2 + 1)/k_2$: for an s -parametrised double series with envelope $|c_{ij}(s)|\rho^{i+j} \leq H(s) \leq C_0(1+s)^D e^{\beta s L}$ and the shifted compatibilities, the merged expansion holds with remainder $O(N^{-\min(p_1 + M_1/k_1, p_2 + M_2/k_2)}(1 + \log N))$ (`twoD_cutoff_general_analytic`), and hence for the chart amplitude $\eta e^{\beta s \xi}$ of two analytic functions (`twoD_cutoff_general_amplitude`). The proof was generated mechanically from the equal-start proof by replacing the three admissibility bounds and the final assembly theorem: the equal-exponent hypothesis entered nowhere else. Zero sorry/axiom.

1 The things formalised

```
theorem corner_gamma_lt_general' (p p : ) (h h k k M M i j : ) (hk : 0 < k)
  (hk : 0 < k) (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k) (hj : j < M) :
  (k : ) * (((h + j : ) : ) + 1) / k - ((h + i : ) : ) - 1
  < ((M + k * M : ) : )
theorem twoD_cutoff_general_analytic ( b p p C L : ) (h h k k M M D : )
  (h : 0 < ) (hb : 0 < b) (hb : b < ) (hk : 0 < k) (hk : 0 < k)
  (hp : ((h : ) + 1) / k = p) (hp : ((h : ) + 1) / k = p)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k)
  (hM : p + ((M : ) - 1) / k < p + (M : ) / k)
  (c :  $\times \rightarrow \rightarrow$ ) (hcc : ij, Continuous (c ij)) (H :  $\rightarrow$ ) (hH : Continuous H)
  (hc : ij s, |c ij s| *  $^{\wedge}$  (ij.1 + ij.2) H s) (hC : 0 C)
  (henv : s, 0 s  $\rightarrow$  H s C * (1 + s)  $^{\wedge}$  D * Real.exp ( * s * L)) :
  (fun N : => twoDAmp b N h h k k (anaAmp c b)
    - mergedSum b h h k k M M (anaFaceU c b) (anaFaceV c b)
      (fun i j s => c (i, j) s) N)
  =0[atTop] fun N : =>
    N  $^{\wedge}$  (-(min (p + (M : ) / k) (p + (M : ) / k))) * (1 + Real.log N)
theorem twoD_cutoff_general_amplitude ... (x y :  $\times \rightarrow$ ) (hx : WSummable x)
  (hy : WSummable y) :
  (fun N : => twoDAmp b N h h k k (anaAmp (ampCoeff x y) b)
    - mergedSum b h h k k M M (anaFaceU (ampCoeff x y) b)
      (anaFaceV (ampCoeff x y) b) (fun i j s => ampCoeff x y (i, j) s) N)
  =0[atTop] fun N : =>
    N  $^{\wedge}$  (-(min (p + (M : ) / k) (p + (M : ) / k))) * (1 + Real.log N)
```

2 Mathematics

Nothing new beyond unit 127: the face envelopes $H(s)\rho^{-i-M_2}/(1-b/\rho)$, the mixed remainder constant $C_0\rho^{-M_1-M_2}/(1-b/\rho)^2$ and the **FaceMoments** constructions from the envelope are exactly as in the equal case; only the three admissibility inequalities $\gamma_i < M_2$, $\gamma'_j < M_1$, $\gamma_{ij} < M_1 + k_1M_2$ now come from the shifted compatibilities (the last via $\gamma_{ij} < M_1$, Astra #9).

3 Paper-facing meaning

Together with unit 127 this is the $d = 2$ cutoff theorem of **thm:TaylorTree** for arbitrary (h, k) , under the paper’s analyticity hypothesis alone, in merged normal form. The regrouping by exponent and the “for every T ” statement for unequal starts follow in the next unit with the cutoffs $q = \max(2T, p_1, p_2) + 1$, $M_\ell = \lceil k_\ell(q - p_\ell) \rceil$.

4 Lean notes

- Generating a variant proof by scripted textual substitution with an assertion per replacement is a reliable way to port a 120-line assembly: every site where the old hypothesis was used is forced to appear in the substitution list, and the compiler checks the rest.
- The generated header kept the original odd line break inside the statement (**c (i, / j) s**); harmless, but the one line that grew past 100 columns was the corner admissibility argument.